

# Kobe Guo

646-275-6596 | [kobe.h.guo@gmail.com](mailto:kobe.h.guo@gmail.com) | [linkedin.com/in/kobeguo](https://linkedin.com/in/kobeguo) | [github.com/KobeGuo99](https://github.com/KobeGuo99) | [kobeguo.com](https://kobeguo.com)

## EDUCATION

### University of Illinois Urbana-Champaign

Master of Computer Science in Data Science (MCS-DS)

Champaign, IL

Expected May 2027

- **GPA: 4.0**

- Relevant coursework: Applied Machine Learning, Data Mining, Cloud Computing, Deep Learning

### Vanderbilt University

Bachelor of Science in Computer Science, Minor in Data Science

Nashville, TN

Aug. 2020 – May 2024

- **GPA: 3.42**

- Relevant Coursework: Algorithms, Data Structures, Database Management Systems, Machine Learning

## EXPERIENCE

### Software Engineer Intern

June 2026 – Aug. 2026

Capital One

McLean, VA

- Developed and enhanced **internal self-service tools** using React and Node.js, reducing average task completion time for support workflows by **25%** based on manual step comparison.
- Built and integrated **RESTful APIs**, improving response latency by **20%** through endpoint optimization and reduced redundant data fetching (measured via local profiling and logs).

### Software Engineer Intern

June 2023 – Aug. 2023

Nourish'd

Cape Town, South Africa

- Developed a React Native app for food ordering, **increasing user engagement by 30%**, with a **50% rise in daily active users**.
- Migrated backend from **MySQL to Firebase**, improving scalability and reducing data retrieval times through real-time synchronization, **boosting backend efficiency by 25%**.

## PROJECTS

### CaliForest (PyHealth Open-Source Contribution) | *Python, scikit-learn, NumPy, PyTorch* [Github](#) | PR #999

- Implemented and contributed a **calibrated Random Forest (CaliForest)** model to the PyHealth open-source library; PR reviewed and **successfully merged**.
- Designed calibration pipeline using **out-of-bag predictions and tree-level variance** with inverse-variance weighting; supported **isotonic and logistic regression**.
- Refactored training to a dedicated `fit()` pipeline (vs minibatch forward pass) and validated on **MIMIC-IV ICU data** using **AUROC and Brier score**.

### URL Shortener | *Node.js, Express, Redis, PostgreSQL, React, Docker, AWS EC2*

[Github](#) | [Live Demo](#)

- Built a production URL shortener implementing **cache-aside pattern** with Redis and PostgreSQL, achieving **5,998 req/sec at 50ms p99 latency** under 100 concurrent connections.
- Implemented **Redis-backed rate limiting** (10 req/min per IP) and async click analytics tracking user-agent, IP, and timestamp per redirect.
- Deployed to **AWS EC2** with Docker Compose, pm2 process management, automated startup migrations, and a React/Tailwind frontend served in production.

### Medical Image Segmentation with Deep Learning | *Python, TensorFlow, Keras*

[Github](#) | [View Project](#)

- Trained a U-Net model on the ULS23 challenge dataset of 3D medical images sourced from hospitals, achieving a **0.87 Dice score** on bone datasets, up from a 0.75 baseline.
- Addressed 3D volume memory constraints by slicing volumetric scans into 2D cross-sections, enabling training on full hospital-grade imaging data without GPU memory overflow.
- Improved model robustness through data augmentation including random flipping, rotation, and noise injection across a multi-contributor medical imaging dataset.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, C++, SQL

**Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Redux, TensorFlow, PyTorch, scikit-learn

**Databases & Caching:** PostgreSQL, MySQL, Redis, Firebase

**Cloud & DevOps:** AWS EC2, Docker, Docker Compose, CI/CD